

# Grade 3 Ontario Curriculum Algebra

Math in My Community

# Unit Overview: Math in My Community

**Grade:** Grade 3 | **Strand:** Algebra

**Target Expectations:** C4

**Learning Goal:** Children will apply the 5-step modelling cycle to community-scale situations: pick a situation, ask a math question, collect data, predict, reflect.

| Lesson | Lesson Title                    | Learning Goal                                                          |
|--------|---------------------------------|------------------------------------------------------------------------|
| Day 1  | Math in My Community            | I can describe a community situation and ask a math question about it. |
| Day 2  | Community Data Day              | I can collect data to answer a math question about my community.       |
| Day 3  | Community Predictions           | I can use a rule to predict a community outcome.                       |
| Day 4  | Community Reflections           | I can compare a prediction to actual data and refine my model.         |
| Day 5  | My Community Modelling Capstone | I can model a real-life community situation from my own life.          |

# Lesson 1: Math in My Community

**Goal:** Math is everywhere in our community: parks, hockey rinks, libraries, restaurants.

**Expectation:** C4

**Learning Goal:** "I can describe a community situation and ask a math question about it."

**Materials:** M1\_community\_map, M3\_modelling\_chart, M4\_situation\_cards, M9\_coach\_chart

## 1. Minds On (12 mins): Coach Cara's Community Tour

Coach Cara: Math is everywhere in our community! Where do you see math?  
Coach Cara opens: 'math questions ANSWER WITH NUMBERS!'

Bring children to the carpet. Coach Cara puppet. 'Friends — I'm Coach Cara the Community Captain. Today we see math everywhere in our community.'

## 2. Action (25 mins): Community Question Stations

- Whole group: pick 4 community locations.
- Three stations.
- Stretch challenge at Station C: write a question that needs 2 numbers (e.g., difference between weekday and weekend visitors).
- Discuss math vs descriptive.
- Circulate to record formative evidence on the question tracker.
- STRATEGY COMPARE: Two pairs share questions about the same location (rink, library, store).

## 3. Consolidation (13 mins): Coach Cara's Question Brag

Discuss:

- What does MODEL mean?
- What's a community location with math?
- Math vs descriptive question example?
- Why are math questions important?

Exit: Coach Cara's refrain — class chants: 'Coach Cara's rule: math questions ANSWER WITH NUMBERS!' (3x).

# Lesson 2: Community Data Day

**Goal:** We answer math questions by collecting DATA. Counts, measurements, observations.

**Expectation:** C4

**Learning Goal:** "I can collect data to answer a math question about my community."

**Materials:** M1\_community\_map, M2\_data\_charts, M3\_modelling\_chart, M7\_two\_colour\_counters

## 1. Minds On (12 mins): Coach Cara's Data Day

Coach Cara: How many people use the rink each Saturday? Let's COUNT. Coach Cara opens: 'COUNTS — not GUESSES!' Yesterday's questions are now today's COUNTS.

Bring children to the carpet. Coach Cara + Sport Sam puppets. 'Yesterday we asked QUESTIONS. Today we ANSWER them by collecting DATA.'

## 2. Action (25 mins): Community Data Stations

- Whole group: collect data for 4 community situations.
- Three stations.
- Stretch challenge at Station C: collect data for TWO different days.
- Discuss data quality.
- Circulate to record formative evidence on the data tracker.
- STRATEGY COMPARE: Two pairs share their tally organisation — one used groups of 10, another used 100s rows.

## 3. Consolidation (13 mins): Coach Cara's Data Brag

Discuss:

- What does DATA mean?
- Why collect data?
- What's the difference between counting and measuring?
- What community thing would you count?

Exit: Coach Cara's data refrain — 'Coach Cara COUNTS — she doesn't GUESS!' (3x).

# Lesson 3: Community Predictions

**Goal:** We use rules and data to PREDICT community outcomes.

**Expectation:** C4

**Learning Goal:** "I can use a rule to predict a community outcome."

**Materials:** M1\_community\_map, M2\_data\_charts, M3\_modelling\_chart, M5\_prediction\_chart

## 1. Minds On (12 mins): Coach Cara's Predictions

Coach Cara: 50 visitors at the rink each Saturday. Predict for 4 weeks. Coach Cara opens: 'PER day, PER week, PER YEAR — rate  $\times$  time = TOTAL!'

Bring children to the carpet. Coach Cara + Sport Sam puppets. 'Friends — yesterday we collected DATA. Today we PREDICT what will happen NEXT.'

## 2. Action (25 mins): Community Prediction Stations

- Whole group: predict for 4 community rules.
- Three stations.
- Stretch challenge at Station C: 2-rule prediction (e.g., morning + evening visitors).
- Discuss prediction logic.
- Circulate to record formative evidence on the prediction tracker.
- INVENT-AND-COMPARE: Two pairs share their Station C scaling rules.

## 3. Consolidation (13 mins): Coach Cara's Prediction Brag

Discuss:

- What does PREDICT mean?
- How do we use a rule to predict?
- Can predictions be wrong?
- What's a community prediction you'd make?

Exit: Coach Cara's prediction refrain — 'PER day, PER week, PER YEAR — rate  $\times$  time = TOTAL!' (3x).

# Lesson 4: Community Reflections

**Goal:** We check predictions against reality. Sometimes the model needs refining.

**Expectation:** C4

**Learning Goal:** "I can compare a prediction to actual data and refine my model."

**Materials:** M2\_data\_charts, M3\_modelling\_chart, M5\_prediction\_chart, M7\_two\_colour\_counters

## 1. Minds On (12 mins): Coach Cara's Reflection

Coach Cara predicted 200 visitors. We counted 185. What does that mean?

Bring children to the carpet. Coach Cara + Sport Sam puppets. 'Yesterday we PREDICTED. Today we REFLECT and REFINE.'

## 2. Action (25 mins): Community Reflection Stations

- Whole group: compare 4 predict-actual pairs.
- Three stations.
- Stretch challenge at Station C: refine TWO different rules in the same situation.
- Discuss the cycle.
- Circulate to record formative evidence on the reflection tracker.

## 3. Consolidation (13 mins): Coach Cara's Reflection Brag

Discuss:

- What does REFINE mean?
- Why don't predictions always match?
- How does refining make the model better?
- What real-life rule would you refine?

Exit: Each child writes ONE refinement on a sticky note: 'I'd refine my rule to \_\_\_\_.'

# Lesson 5: My Community Modelling Capstone

**Goal:** We can model ANY community situation. Pick a topic, ask a question, build a model.

**Expectation:** C4

**Learning Goal:** "I can model a real-life community situation from my own life."

**Materials:** M1\_community\_map, M4\_situation\_cards, M6\_capstone\_template, M3\_modelling\_chart

## 1. Minds On (12 mins): Coach Cara's Final Showcase

Now you model YOUR community. Pick a situation. Walk through all 5 steps.

Bring children to the carpet. Coach Cara + Sport Sam together. 'Friends — today you become MATHEMATICAL MODELLERS. Pick a community situation. Walk through the cycle: SITUATION, QUESTION, DATA, PREDICT, REFLECT.'

## 2. Action (25 mins): Capstone Modelling

- Hand out capstone templates.
- Children work for 12-15 minutes individually.
- Stretch challenge: include a refinement based on data.
- Gallery walk: children swap and read each other's models.
- Whole-class share: 5-6 children share aloud.

## 3. Consolidation (13 mins): Coach Cara's Closing Brag and Certificate Ceremony

Discuss:

- What's the most surprising community model you saw today?
- Why do we use the modelling CYCLE?
- What community situation will you model after this unit?
- What's the difference between data and prediction?

Exit: Each child receives a Junior Community Modeller certificate.

Name: \_\_\_\_\_

Date: \_\_\_\_\_

# Math in My Community

**Learning Goal:** I can describe a community situation and ask a math question about it.

## Instructions for the Student:

1. Pick a community location.
2. Write 4 math questions about your community.
3. Sort 6 questions into MATH and NOT MATH.

## My Questions

List 3 questions to investigate:

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Name: \_\_\_\_\_

Date: \_\_\_\_\_

# Coach Cara's Community Data

**Learning Goal:** I can collect data to answer a math question about my community.

## Instructions for the Student:

1. Tally and record data for 4 community situations.
2. Use your data to answer 3 questions.
3. Pick something to count in your community.

## Use the Data

Use the data to answer 3 questions:

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|  |

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Name: \_\_\_\_\_ Date: \_\_\_\_\_

## Coach Cara's Predictions

**Learning Goal:** I can use a rule to predict a community outcome.

### Instructions for the Student:

1. Rule: 50 visitors per Saturday.
2. Use 4 community rules.
3. Pick your own community rule.

### Part 3: My Community Rule

**Pick your own community rule. Predict for 12 weeks.**

[illegible]

Name: \_\_\_\_\_

Date: \_\_\_\_\_

# Coach Cara's Reflections

**Learning Goal:** I can compare a prediction to actual data and refine my model.

## Instructions for the Student:

1. Compare 5 predict-vs-actual scenarios.
2. Refine 3 rules based on actual data.
3. Reflect on a real community prediction.

## Refine Your Prediction

**Original prediction:**

**What I noticed:**

**New prediction:**

Name: \_\_\_\_\_

Date: \_\_\_\_\_

# My Community Model

**Learning Goal:** I can model a real-life community situation from my own life.

## Instructions for the Student:

1. Pick a community topic.
2. Write a math question about your situation.
3. Collect or imagine data.

## My Topic

Pick a real-life topic:

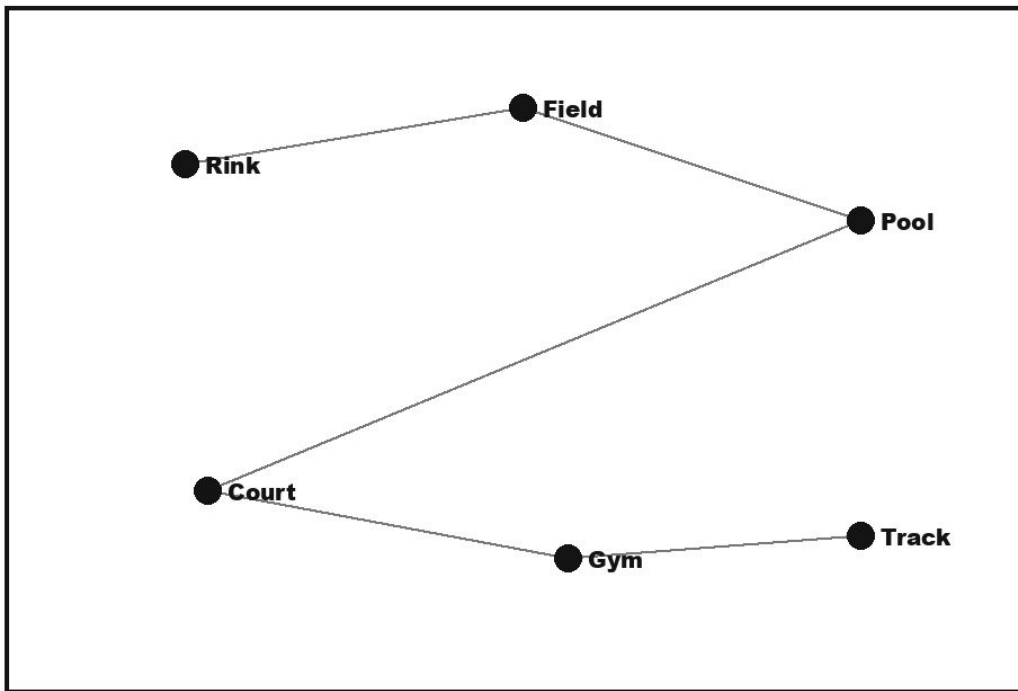
|  |
|--|
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|  |

# Paper Community Map

(M1\_community\_map)

Letter-size map showing community locations.

## Community Map



### PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

### TEACHER PREP

1. Print 1 per pair.

2. Laminate.

3. Hand out Lesson 1.

4. Reuse Lessons 2-5.

5. Optional extras.

### USE IN CLASS

Quantity: 1 per pair plus 1 anchor

Lessons: 1, 2, 3, 5

Prep: ~12 min

# Community Data Charts

(M2\_data\_charts)

6 chart templates for recording community-scale counts.

## Data Chart

| Day | Count | Notes |
|-----|-------|-------|
|     |       |       |
|     |       |       |
|     |       |       |
|     |       |       |
|     |       |       |
|     |       |       |

### PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

### TEACHER PREP

1. Print 1 page per pair.

2. Cut into 6 charts.

3. Laminate.

4. Hand out Lesson 2.

5. Reuse Lessons 3-4.

### USE IN CLASS

Quantity: 6 per pair

Lessons: 2, 3, 4

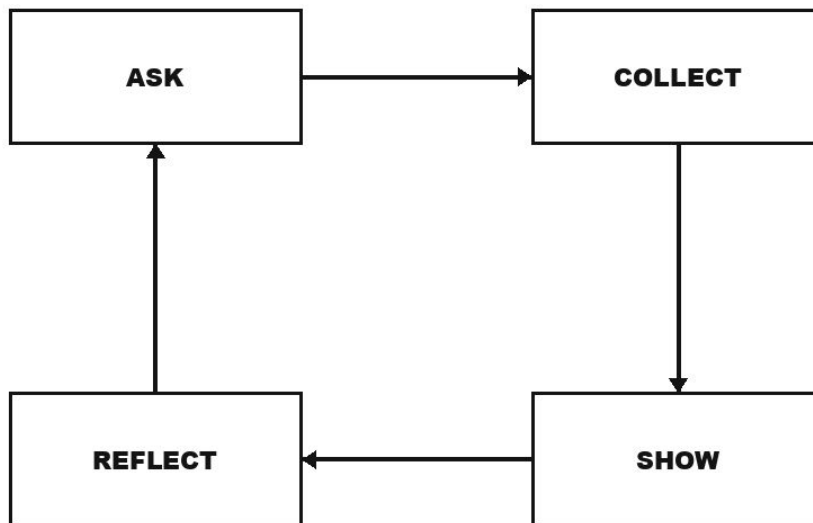
Prep: ~12 min

# Modelling Cycle Chart

(M3\_modelling\_chart)

Letter-size chart showing the 5 modelling steps with looping arrows.

## Modelling Cycle



### PRINT SPECS

Size: letter landscape  
Colour: bw  
Pages per set: 1  
Laminate: yes

### TEACHER PREP

1. Print 1 chart per pair.
2. Laminate.
3. Hand out Lesson 1.
4. Reference all week.
5. Reuse.

### USE IN CLASS

Quantity: 1 per pair plus 1 anchor  
Lessons: 1, 2, 3, 4, 5  
Prep: ~12 min

# Community Situation Cards

(M4\_situation\_cards)

12 community situation cards.

## Situation Cards

|                                                   |                                      |
|---------------------------------------------------|--------------------------------------|
| Counted 12 trees on Day 1.                        | Tapped 8 trees on Day 2.             |
| Filled 5 buckets on Day 3.                        | Took 3 days to fill the jar.         |
| Friend says the syrup tastes sweetest at the end. | Predict: how many buckets next week? |

### PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

### USE IN CLASS

Quantity: 12 cards

Lessons: 1, 5

Prep: ~15 min

### TEACHER PREP

1. Print 2 pages on cardstock.

2. Laminate.

3. Cut into 12 cards.

4. Sort by topic.

5. Distribute Lessons 1 and 5.



# Prediction Chart

(M5\_prediction\_chart)

Letter-size chart with PREDICT, ACTUAL, GAP columns.

## Predict vs. Actual

| Day | Predicted | Actual | Difference |
|-----|-----------|--------|------------|
|     |           |        |            |
|     |           |        |            |
|     |           |        |            |
|     |           |        |            |
|     |           |        |            |
|     |           |        |            |

### PRINT SPECS

Size: letter landscape  
Colour: bw  
Pages per set: 1  
Laminate: yes

### USE IN CLASS

Quantity: 1 per pair  
Lessons: 3, 4  
Prep: ~12 min

### TEACHER PREP

1. Print 1 chart per pair.
2. Laminate.
3. Hand out Lesson 3.
4. Use dry-erase.
5. Reuse Lesson 4.

# Capstone Modelling Template

(M6\_capstone\_template)

Letter-size template for Lesson 5.

| Capstone Modelling Template |                           |
|-----------------------------|---------------------------|
| 1                           | <b>Situation</b><br>_____ |
| 2                           | <b>Question</b><br>_____  |
| 3                           | <b>Data</b><br>_____      |
| 4                           | <b>Predict</b><br>_____   |

## PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

## TEACHER PREP

1. Print 1 per child.

2. Hand out Lesson 5.

3. Children fill in.

4. Collect for assessment.

5. Display.

## USE IN CLASS

Quantity: 1 per child

Lessons: 5

Prep: ~12 min

# Two-Colour Counters

(M7\_two\_colour\_counters)

30 counters per pair for tallying community-scale data.

## Counter Sheet

|           |           |           |           |           |
|-----------|-----------|-----------|-----------|-----------|
| <b>1</b>  | <b>2</b>  | <b>3</b>  | <b>4</b>  | <b>5</b>  |
| <b>6</b>  | <b>7</b>  | <b>8</b>  | <b>9</b>  | <b>10</b> |
| <b>11</b> | <b>12</b> | <b>13</b> | <b>14</b> | <b>15</b> |
| <b>16</b> | <b>17</b> | <b>18</b> | <b>19</b> | <b>20</b> |

### PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

### TEACHER PREP

1. Print 1 per pair.

2. Laminate.

3. Cut into 30 counters.

4. Sort.

5. Distribute Lessons 2-4.

### USE IN CLASS

Quantity: 30 per pair

Lessons: 2, 3, 4

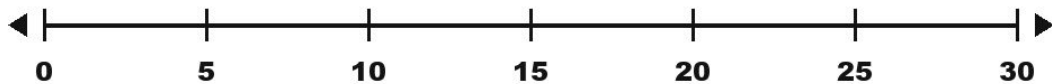
Prep: ~15 min

# Community Number Line to 1000

(M8\_number\_line)

Letter-size number line marked 0 to 1000.

## Number Line



### PRINT SPECS

Size: letter landscape  
Colour: bw  
Pages per set: 1  
Laminate: yes

### TEACHER PREP

1. Print 1 per pair.
2. Laminate.
3. Hand out Lesson 2.
4. Reuse Lessons 3-4.

### USE IN CLASS

Quantity: 1 per pair  
Lessons: 2, 3, 4  
Prep: ~10 min

# Coach's Modelling Anchor Chart

(M9\_coach\_chart)

Anchor chart with community examples.

| Coach Cara's Stat Chart                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Goals scored</li><li>• Saves made</li><li>• Team practice days</li><li>• Equipment shared</li></ul> |

## PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

## TEACHER PREP

1. Print 4 pages and tile.

2. Laminate.

3. Tape to wall.

4. Reference all week.

5. Roll for re-use.

## USE IN CLASS

Quantity: 1 chart

Lessons: 1, 2, 3, 4

Prep: ~18 min

# Math in My Community Anchor

(M10\_community\_poster)

Anchor poster of Coach Cara and Sport Sam.

## Math in My Community

- Track stats. Plan events.
- Sort by category.
- Tally + total.
- Show on a graph.
- Improve your community.

### PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

### USE IN CLASS

Quantity: 1 poster

Lessons: 1, 2, 3, 4, 5

Prep: ~18 min

### TEACHER PREP

1. Print 4 pages and tile.

2. Laminate.

3. Tape to wall.

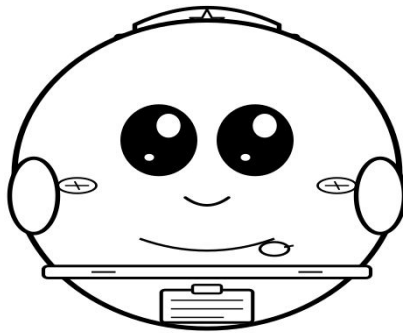
4. Reference all week.

5. Roll for re-use.

# Coach Cara Character Puppet

(char\_cara\_puppet)

Printable Cara illustration on a popsicle stick.



**Coach Cara**

## PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

## USE IN CLASS

Quantity: 1 puppet

Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

## TEACHER PREP

1. Print on cardstock.

2. Cut.

3. Glue popsicle stick.

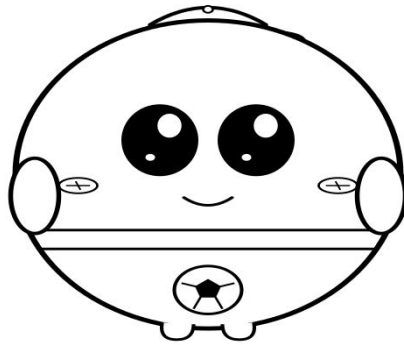
4. Optional: laminate.

5. Store flat.

# Sport Sam Character Puppet

(char\_sam\_puppet)

Printable Sam illustration on a popsicle stick.



**Sport Sam**

## PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

## USE IN CLASS

Quantity: 1 puppet

Lessons: 2, 3, 4, 5

Prep: ~10 min

## TEACHER PREP

1. Print on cardstock.

2. Cut.

3. Glue popsicle stick.

4. Optional: laminate.

5. Store flat.



Name: \_\_\_\_\_

Date: \_\_\_\_\_

# Summative Assessment Rubric

| Expectation                                                                                                                                     | Level 1<br>(Beginning)                                        | Level 2<br>(Developing)                                   | Level 3<br>(Achieving)                                                             | Level 4<br>(Exceeding)                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>C4</b><br>apply the process of mathematical modelling to represent, analyse, make predictions, and provide insight into real-life situations | Cannot identify a community situation or ask a math question. | With prompts, walks through 2-3 of the 5 modelling steps. | Independently walks through all 5 modelling steps with a coherent community topic. | Walks through all 5 steps with community-scale numbers and a refinement based on the data. |

# Junior Community Modeller Certificate

This certificate is proudly presented to:

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For successfully completing the 5-Day Math in My Community unit and demonstrating the skills of a Junior Community Modeller. Coach Cara and Sport Sam congratulate the student.

**By completing this unit, this student has demonstrated:**

- Describing a community situation
- Asking a math question
- Collecting community-scale data
- Making a community-scale prediction
- Reflecting and refining the model

# Marketplace Listing

A 5-day Ontario Grade-3 unit on mathematical modelling (C4) framed around 'Math in My Community' — Coach Cara the Community Captain and Sport Sam the Sports Reporter.

Price: \$9.99 CAD

Pages: ~32

Classroom time: 250 min

Teacher prep: ~90 min

Top tags: grade-3, math, ontario-curriculum, modelling, algebra, real-life, canadian, community

## What's included:

- 5 fully-scripted lesson plans with 50-minute durations and 250 minutes of total teaching time
- 5 student worksheets aligned to each lesson
- 1 mid-unit formative assessment (Community Modelling Check-In)
- 10 printable teacher manipulatives plus 2 character puppets: community map, data charts (6), modelling cycle chart, community situation cards (12), prediction...
- 1 end-of-unit reflection sheet with 5 self-assessment prompts
- 4 class observation trackers: 1 diagnostic + 3 formative
- 1 single-row summative rubric covering C4 across 4 levels
- 1 summative-administration guide tying the rubric to Lesson 5 capstone
- Junior Community Modeller certificate (one per child)
- Detailed expanded walkthroughs covering common misconceptions
- Differentiation tracks built into every lesson